

Research Proposal

SLEEP: Synaptic Learning through Error-driven Encoding and Plasticity

A Metabolic Memory Framework for Continuous Learning

The Core Thesis

Large Language Models suffer from **Architectural Anterograde Amnesia**—the complete inability to form new persistent memories after training. The field's solution has been **Retrieval-Augmented Generation (RAG)**: give the amnesiac a notebook to frantically search through.

This is a prosthetic. Not a cure.

We propose something different: **Restore the hippocampus. Teach the model to sleep.**

Biological memory does not work through lookup tables. It works through three mechanisms that ML has largely ignored:

1. **Synaptic Tagging**: Experiences leave temporary "marks" on synapses—cheap, fast, noncommittal
2. **Protein Synthesis**: Only important marks get stabilized by expensive molecular machinery
3. **Sleep Replay**: During rest, important memories are replayed to permanent storage, compressed into gist

The result is a system that **learns instantly, forgets gracefully, and consolidates automatically**—all while managing its own memory without external intervention.

We build this for language models.

The Architecture: The "Living Memory"

We propose a three-system architecture based on the biology of memory consolidation.

1. The Tagging Layer — "The Noticing"

Biological Analogue: Synaptic Tagging (Frey & Morris, 1997)

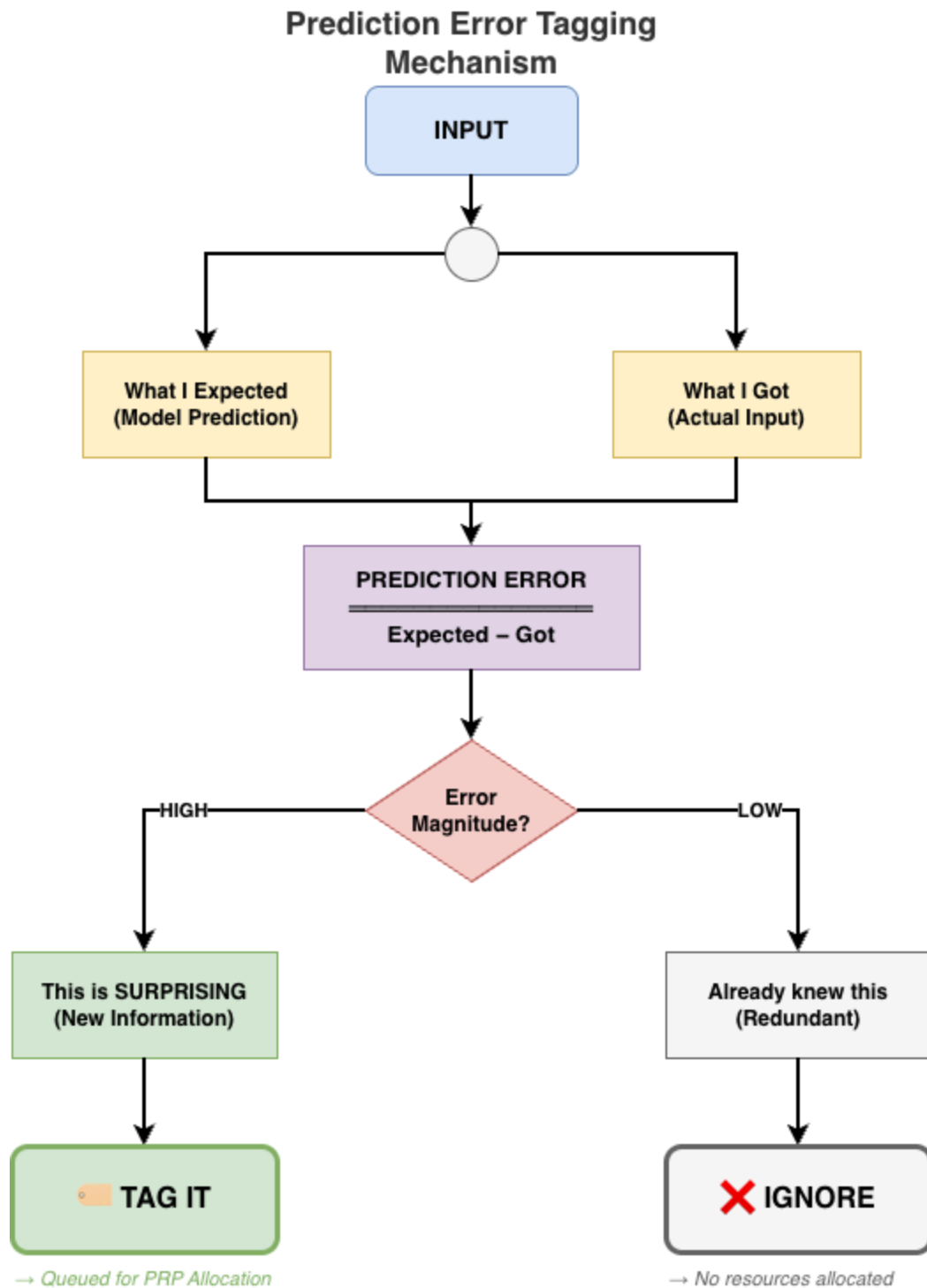
The Biology:

When you experience something, your synapses don't immediately decide "keep this forever" or "forget this." Instead, they create a **tag**—a temporary molecular marker that says "something happened here." Tags are cheap. Tags are fast. Tags require no commitment.

But tags decay within hours. Unless something else happens.

The Mechanism:

When the model ingests new information, it doesn't store the information directly. It computes **prediction error**—the gap between what the model expected and what it actually received.



The tags are sparse. They are pointers, not storage. They mark *where* surprising information lives, not the information itself.

Why Prediction Error?

Approach	What Gets Tagged
Store everything	Everything (no prioritization)
Hebbian (correlation)	Frequent patterns (not necessarily useful)
Prediction Error	Surprising things (genuinely new information)

Result: The document is not "memorized." It is **indexed by surprise**. The model knows where the new information is and what about it was unexpected. This costs almost nothing.

2. The Protein Budget — "The Prioritizing"

Biological Analogue: Plasticity-Related Proteins (PRPs) and Metabolic Constraints

The Biology:

Tags alone decay within hours. For a memory to persist, the tagged synapse must "capture" **Plasticity-Related Proteins**—molecular machinery that stabilizes the connection.

Here is the critical insight: **PRPs are expensive**. The brain cannot afford to stabilize every memory. It has a limited metabolic budget. PRPs must be allocated strategically.

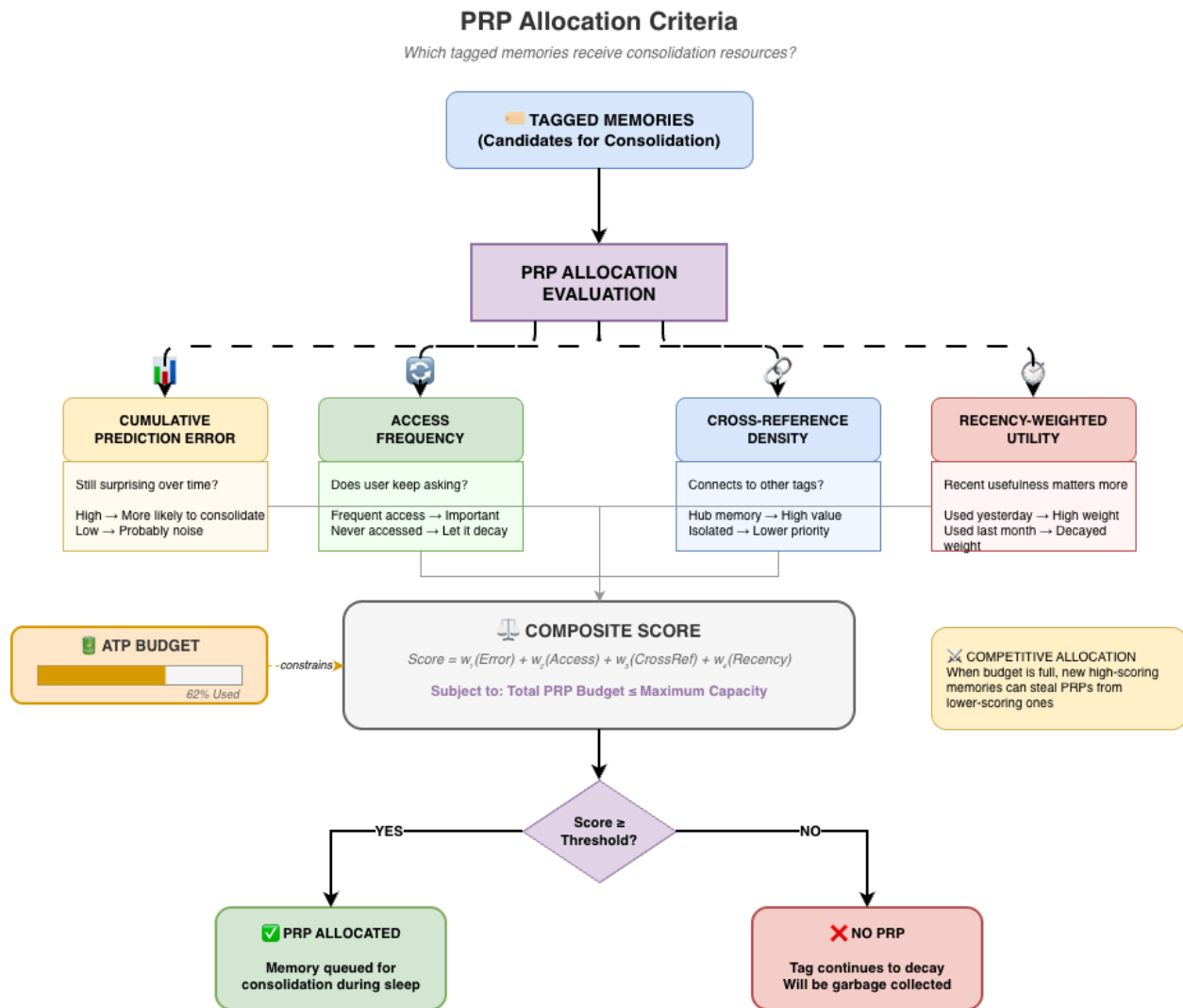
What determines allocation?

- **Novelty:** Is this still surprising after repeated exposure?
- **Behavioral Relevance:** Did this information prove useful?
- **Emotional Salience:** Does this matter to the organism?

Tags compete for a limited pool of PRPs. Winners get stabilized. Losers decay.

The Mechanism:

The model maintains a bounded **Protein Budget**—a hard limit on how many tagged memories can be promoted to consolidation candidates.



The Dynamics:

Tag Status	What Happens
Tagged, unaccessed	Decays naturally over time
Tagged, accessed occasionally	Persists but doesn't consolidate
Tagged, accessed frequently	Receives PRP allocation
Tagged + PRP	Queued for consolidation
Consolidated	Tag cleared, memory now permanent

The Metabolic Constraint:

The total PRP budget is fixed. If new, more important memories arrive, they can **steal** PRPs from less important ones. This is competitive consolidation. The system self-organizes around what actually matters.

Result: The model cannot remember everything. It must prioritize. The prioritization is automatic, based on prediction error and access patterns. No external memory management required.

3. The Sleep Engine — "The Becoming"

Biological Analogue: Sleep-Dependent Memory Consolidation and Hippocampal Replay

The Biology:

During sleep, something remarkable happens. The hippocampus "replays" the day's important experiences to the neocortex—often compressed, often abstracted, often combined with old memories. Through this replay, temporary hippocampal traces become permanent neocortical knowledge.

Crucially, replay is **generative**. The hippocampus doesn't play back a recording. It reconstructs the experience, filling in gaps, extracting gist, integrating with existing knowledge. You remember that you went to a restaurant. You don't remember every word spoken.

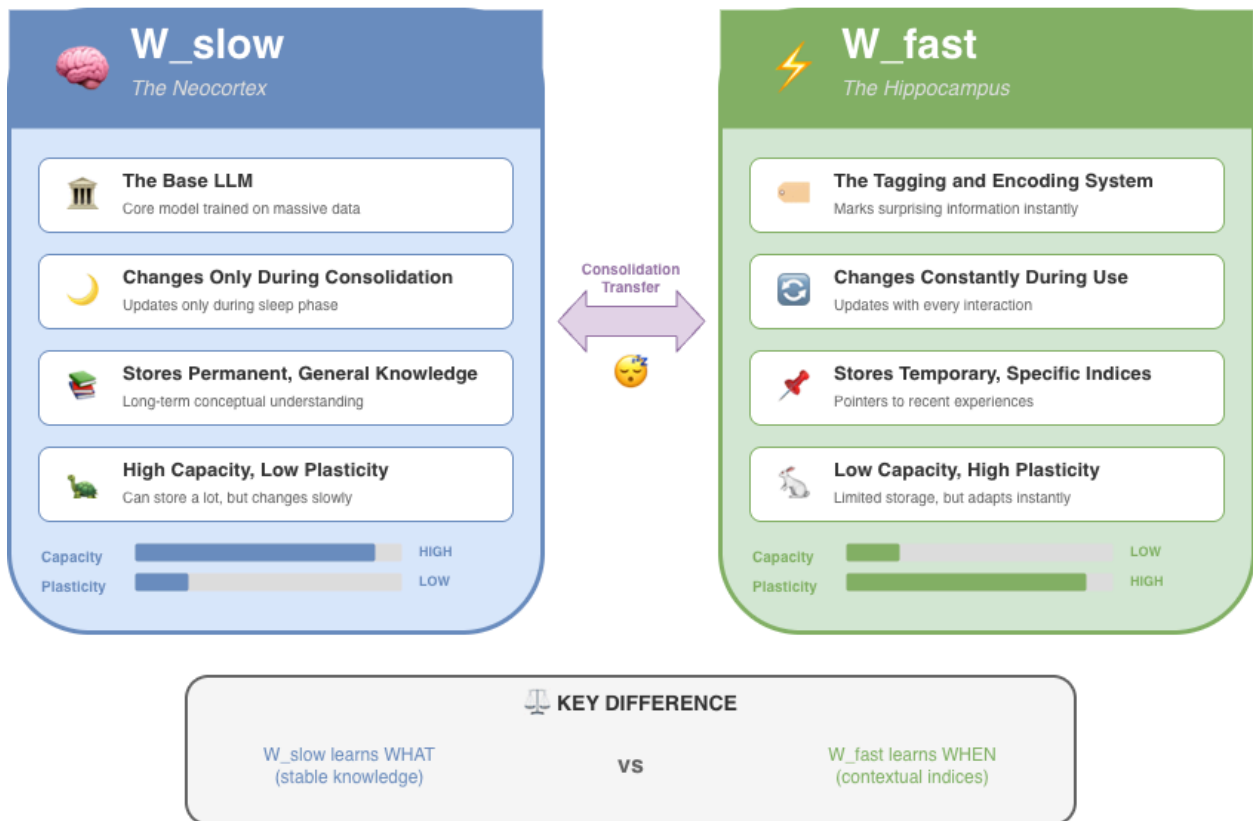
This is why sleep deprivation destroys memory consolidation. No replay, no permanence.

The Mechanism:

The model has two weight systems:

The Dual Weight System

Biologically-Inspired Memory Architecture

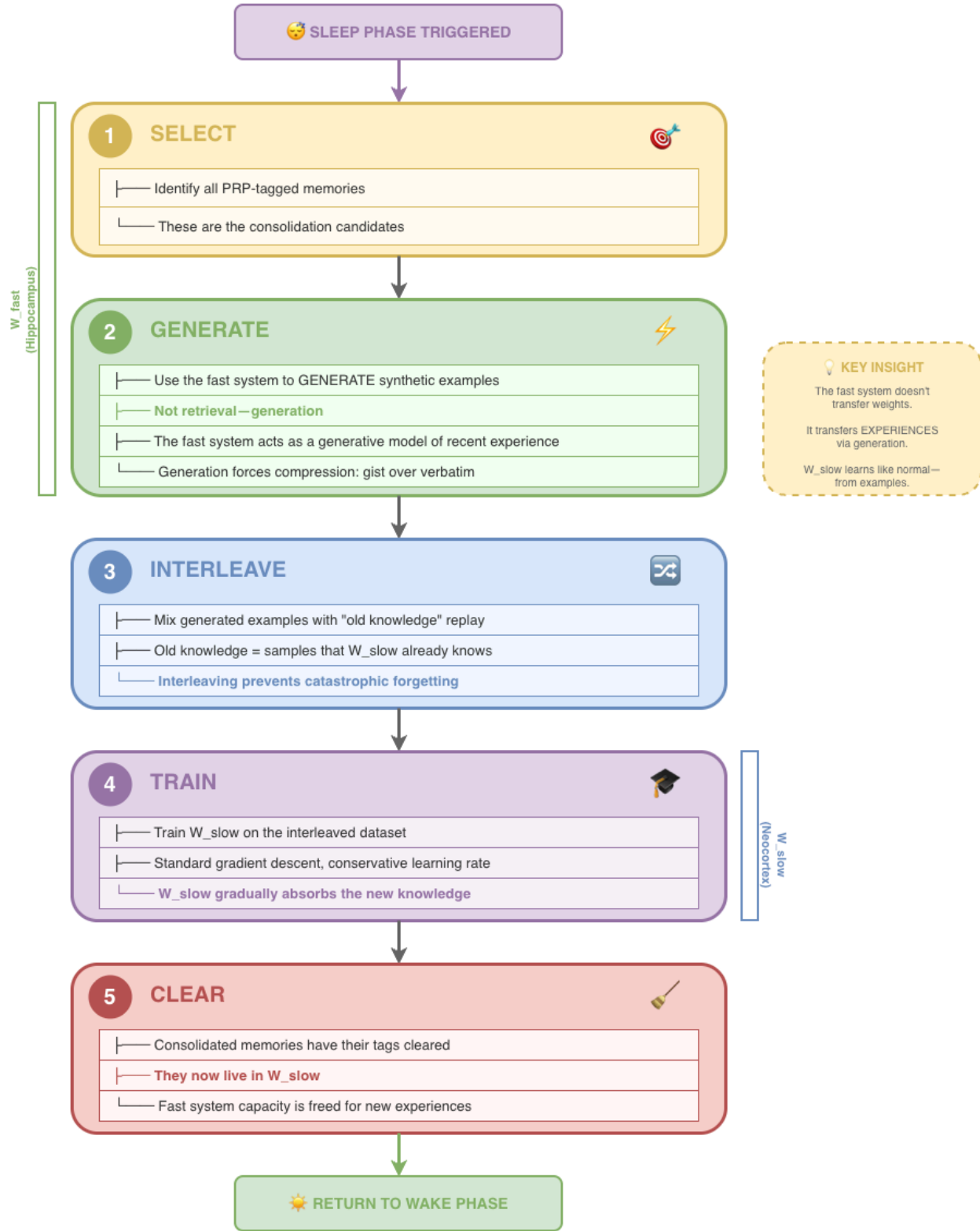


The Consolidation Process:

When triggered (idle time, schedule, or memory pressure), the system enters **Sleep Phase:**

THE SLEEP CONSOLIDATION PROCESS

How memories transfer from W_{fast} to W_{slow}



Why Generation, Not Transfer?

Approach	Problem
Copy weights directly	Architectures may not align. No compression. Interference.
Distillation via logits	Requires paired inputs. Limited abstraction.
Generative replay	Forces compression. Natural abstraction. Unlimited examples.

The Compression Requirement:

Because the fast system generates (not retrieves), it cannot reproduce verbatim. It must compress:

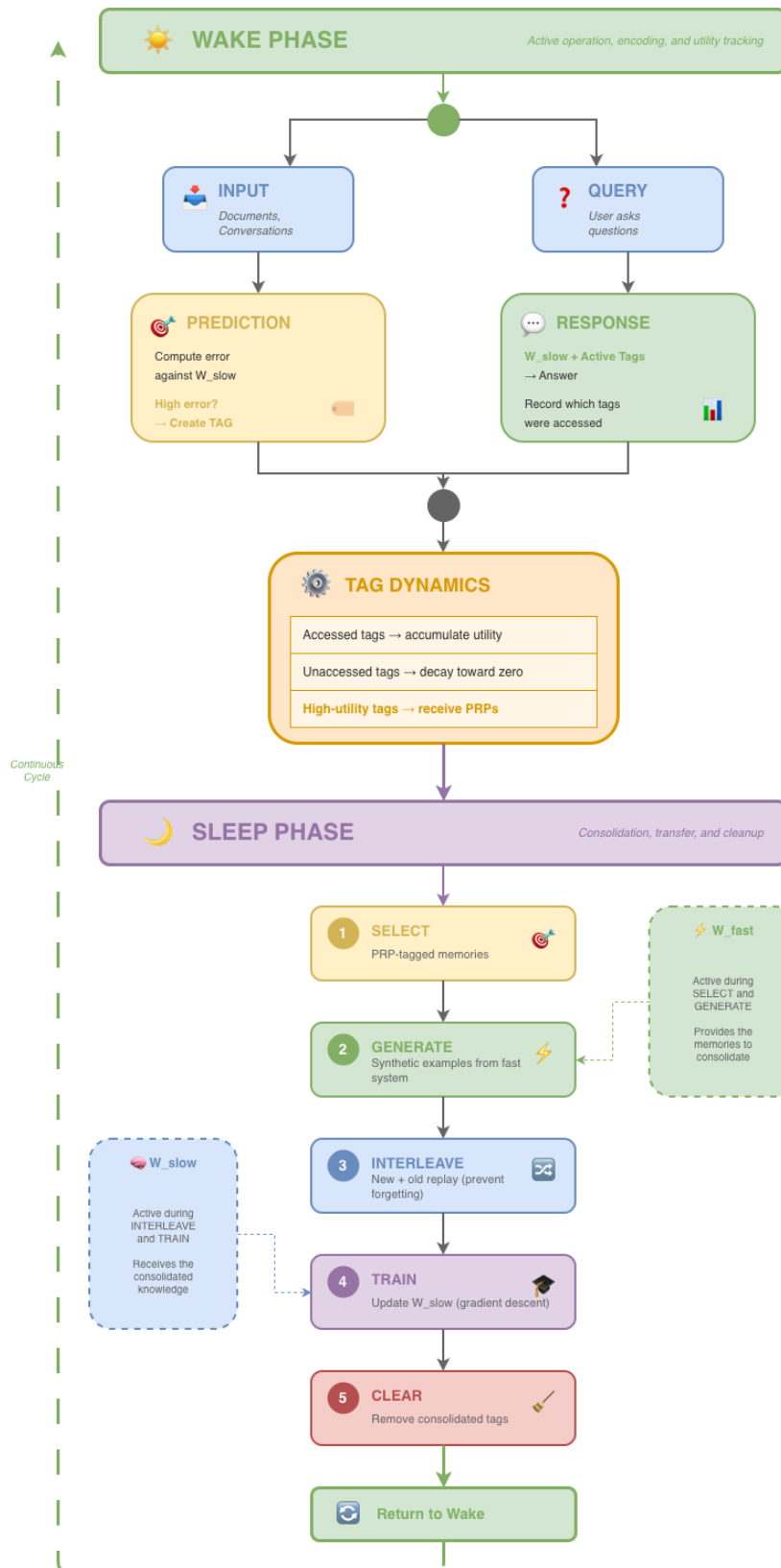
Fast System Encoding	Generated Replay	W_slow Learns
"User asked about Python errors 47 times with specific stack traces"	Synthetic examples of Python debugging conversations	"User is learning Python, prefers debugging help"
"Document contains Q3 revenue of \$4.2M, down 12% from Q2"	Generated Q&A about Q3 financial performance	"Q3 had revenue challenges"
"User corrected my code style 12 times"	Synthetic examples with corrected style	Adjusted code generation patterns

Result: The model develops genuine long-term learning. Not memorization—understanding. The kind that transfers, generalizes, and persists.

The Complete System

SLEEP: Complete System Architecture

Synaptic Learning through Error-driven Encoding and Plasticity



Why This Replaces RAG

Limitation of RAG	How Sleep Architecture Solves It
Retrieval failures (wrong chunks retrieved)	No retrieval—knowledge is in the weights
Context fragmentation (chunks lose coherence)	Generative replay preserves relational structure
No true integration (pattern matching, not understanding)	Training on replay creates real weight changes
Static (never learns from use)	Continuous consolidation improves the model
Hard context limits	Graceful decay manages capacity automatically
Manual memory management	Self-organizing via prediction error + access patterns
No personalization	Long-term user patterns consolidate into W_{slow}

The Research Question

Can a language model with synaptic tagging (prediction-error-driven memory marking), metabolic PRP budgets (bounded consolidation resources), and generative sleep replay (compression-based transfer to permanent weights) achieve continuous, interference-free learning—replacing external retrieval with internal memory?

Sub-Questions

1. **Tagging Validity:** Does prediction error reliably identify information that will be useful for future queries?
2. **Budget Dynamics:** How does PRP budget size affect the quality/coverage tradeoff in consolidation?
3. **Replay Fidelity:** Can generative replay preserve enough information for effective downstream use while achieving sufficient compression?

- 4. **Interference Prevention:** Does interleaved replay successfully prevent catastrophic forgetting in W_slow?
- 5. **Emergent Prioritization:** Does competitive PRP allocation produce prioritization that matches human judgments of importance?
- 6. **Sleep Scheduling:** What triggers optimal consolidation—idle time, memory pressure, or scheduled intervals?

What Success Looks Like

A model that:

- **Reads a document once** and knows it—no retrieval required
- **Forgets what you stop caring about**—automatically, gracefully
- **Remembers what matters**—without being explicitly told what matters
- **Develops genuine expertise** over long-term use—not just access to information
- **Sleeps to consolidate**—turning temporary exposure into permanent capability
- **Never hits a context wall**—capacity is managed metabolically, not architecturally
- **Improves with every conversation**—because every conversation is potential learning

Biological Grounding

This is not metaphor. Each component maps to established neuroscience:

Architecture Component	Biological Mechanism	Key Citation
Tagging Layer	Synaptic Tagging	Frey & Morris, 1997
Prediction Error Signal	Predictive Coding	Rao & Ballard, 1999
PRP Budget	Metabolic Constraints on Plasticity	Redondo & Morris, 2011
Competitive Consolidation	Synaptic Tagging and Capture	Morris, 2006

Architecture Component	Biological Mechanism	Key Citation
Sleep Replay	Hippocampal Sharp-Wave Ripples	Wilson & McNaughton, 1994
Interleaved Training	Complementary Learning Systems	McClelland et al., 1995
Generative Reconstruction	Memory Reconsolidation	Nader et al., 2000

We are not inventing a memory system. We are implementing the one biology already discovered.

Differentiation From Prior Work

The synthesis is novel: Synaptic Tagging + Predictive Saliency + Metabolic Budgets + Generative Sleep Replay has not been combined before.

Prior Work	Their Focus	Our Contribution
Fast Weight Programmers (Schmidhuber, 1992)	Temporary weight storage	We add tagging, PRPs, and principled consolidation
Continual Learning (EWC, PackNet, etc.)	Preventing forgetting	We add biological encoding mechanism and sleep
Generative Replay (Shin et al., 2017)	Replay for continual learning	We add synaptic tagging for selective replay
Memory Networks (Weston, 2014)	External differentiable memory	We modify internal weights, no external storage
RAG (Lewis et al., 2020)	External retrieval	We eliminate retrieval entirely
Predictive Coding in ML (Millidge, 2021)	Efficient learning	We use for saliency detection, not training

Open Challenges (Acknowledged)

We are proposing research, not a solved system. Key challenges:

Challenge	The Hard Part
What is a "tag" computationally?	Sparse mask? Attention pattern? Memory address?

Challenge	The Hard Part
What is a "PRP" computationally?	Learning rate multiplier? Consolidation flag? Weight allocation?
How does the fast system generate?	VAE? Diffusion model? Autoregressive?
How to scale to billions of parameters?	Sparse operations? Layer-specific tagging?
How to validate "correct" consolidation?	What benchmarks measure this?

These define the research. They are not obstacles—they are the work.

Conclusion

For years, we have treated language model memory as a retrieval problem. We gave amnesiacs notebooks and called it solved.

But memory is not retrieval. Memory is encoding, prioritization, decay, and consolidation. Memory is metabolic. Memory requires sleep.

We propose building language models that don't just access information—they **learn** it, **forget** what doesn't matter, and **sleep** to make the important things permanent.

The notebook was always a workaround.

It's time to teach machines to dream.

This document represents a research direction. Mathematical formalization and empirical validation are the next phases of this work.